AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 35 · TEACHER GUIDE

Capstone limitations and revision

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make honest failure analysis part of a successful explanation.

Use checked residuals, subset denominators and fresh-case guidance; preserve original evaluation history.

Keep one habit: Model a bounded result → diagnose architecture limits → assess new frozen cases → plan a properly evaluated revision.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Evaluate a frozen candidate and baseline on final rows

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator; optional local Python 3 interpreter

Fictional data only; reserve the independent task and fresh retry until assessment

No. Once final results influence changes, those rows become development **evidence** for the revised procedure.

Opening retrieval

Without opening the old lesson, explain one key idea from Capstone controlled experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Trace matched two-update learning-rate experiments

Reference example, not the only acceptable wording: Teacher explanation: Hold **training** and validation rows, initial a, architecture/bias policy, loss, update count and evaluation metric fixed. Change learning rate.

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

Look for: Record a network experiment with enough detail to replay its calculations

Reference example, not the only acceptable wording: Teacher explanation: Predict max(0, a*x); start a = 1 with no bias; rate 0.1; two full-batch updates; MSE is mean squared **prediction** error.

T1 and T2 train, V1 validates; no shuffling or normalisation. Select lower validation MSE, ties to earlier update. Record the actual Python version, alongside the code and Python float arithmetic.

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a predicted class, confidence score and observed correctness

Reference example, not the only acceptable wording: Teacher explanation: B predicts 1 with confidence 0.90 and is wrong. E predicts 0 with confidence 0.55 and is correct. F predicts 0 with confidence 0.90 and is wrong.

F’s low score for class 1 corresponds to a high score for class 0, the class selected.

Pictured end goal

Explain capstone limitations and **revision** with a traceable toy-network case and a bounded conclusion.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Capstone controlled experiments

Documenting a network experiment

Uncertainty and failure analysis

For the compact alternative schedule, use two 45-minute sessions. This capstone uses constructed scalar ReLU regression so every calculation is inspectable. It is an experiment in **training** and **evidence**, not a real-world deployment.

Preserve written plans, data roles and raw **prediction**s. Optional local Python requires no external packages or AI account. Do not expose independent-task solutions before collecting work.

Outcomes

Evaluate a frozen candidate and baseline on final rows

Identify a distribution limitation from raw errors

Distinguish **revision** checks from new independent evaluation

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach improvement and failure as compatible findings

> Keep the frozen comparison separate from proposed signed-output checks.

Frozen main procedure and complete prediction rows

Freeze the Week 34 candidate: max(0, a*x), initial a = 1, rate 0.2, two full-batch updates on (1,2),(2,4), no bias. The final weight is 2. The baseline is unchanged at a = 1.

Now open final rows F1=(5,10), F2=(6,12), F3=(-1,-2), F4=(-2,-4). These fictional rows deliberately include a condition absent from the positive **training** and validation **input**s.

Opening them is an evaluation event, not permission to tune on them while retaining an independent-test claim.

Write raw **prediction**s for both procedures before summarising. A low overall error can hide a systematic problem in a condition. These four constructed examples do not estimate prevalence or real-world performance.

ID · x · y · Baseline prediction · Candidate prediction

> **ID:** F1
> **x:** 5
> **y:** 10
> **Baseline prediction:** 5
> **Candidate prediction:** 10

> **ID:** F2
> **x:** 6
> **y:** 12
> **Baseline prediction:** 6
> **Candidate prediction:** 12

> **ID:** F3
> **x:** -1

> **y:** -2
>
> **Baseline prediction:** 0
>
> **Candidate prediction:** 0

> **ID:** F4
>
> **x:** -2
>
> **y:** -4
>
> **Baseline prediction:** 0
>
> **Candidate prediction:** 0

Checked overall and subset arithmetic

Baseline squared errors are 25,36,4,16, so MSE is 81/4 = 20.25. Candidate squared errors are 0,0,4,16, so MSE is 20/4 = 5. The candidate passes the strict lower-MSE final criterion on these four rows.

On the two negative-**input** rows its MSE is (4+16)/2 = 10; on the positive rows it is zero.

Final case · Baseline residual · Baseline squared error · Candidate residual · Candidate squared error

> **Final case:** F1
>
> **Baseline residual:** -5
>
> **Baseline squared error:** 25
>
> **Candidate residual:** 0
>
> **Candidate squared error:** 0

> **Final case:** F2
>
> **Baseline residual:** -6
>
> **Baseline squared error:** 36
>
> **Candidate residual:** 0
>
> **Candidate squared error:** 0

Final case: F3

Baseline residual: 2

Baseline squared error: 4

Candidate residual: 2

Candidate squared error: 4

Final case: F4

Baseline residual: 4

Baseline squared error: 16

Candidate residual: 4

Candidate squared error: 16

Final case: All four rows

Baseline residual: Total squares 81

Baseline squared error: MSE 81/4 = 20.25

Candidate residual: Total squares 20

Candidate squared error: MSE 20/4 = 5

Final case: Candidate positive subset

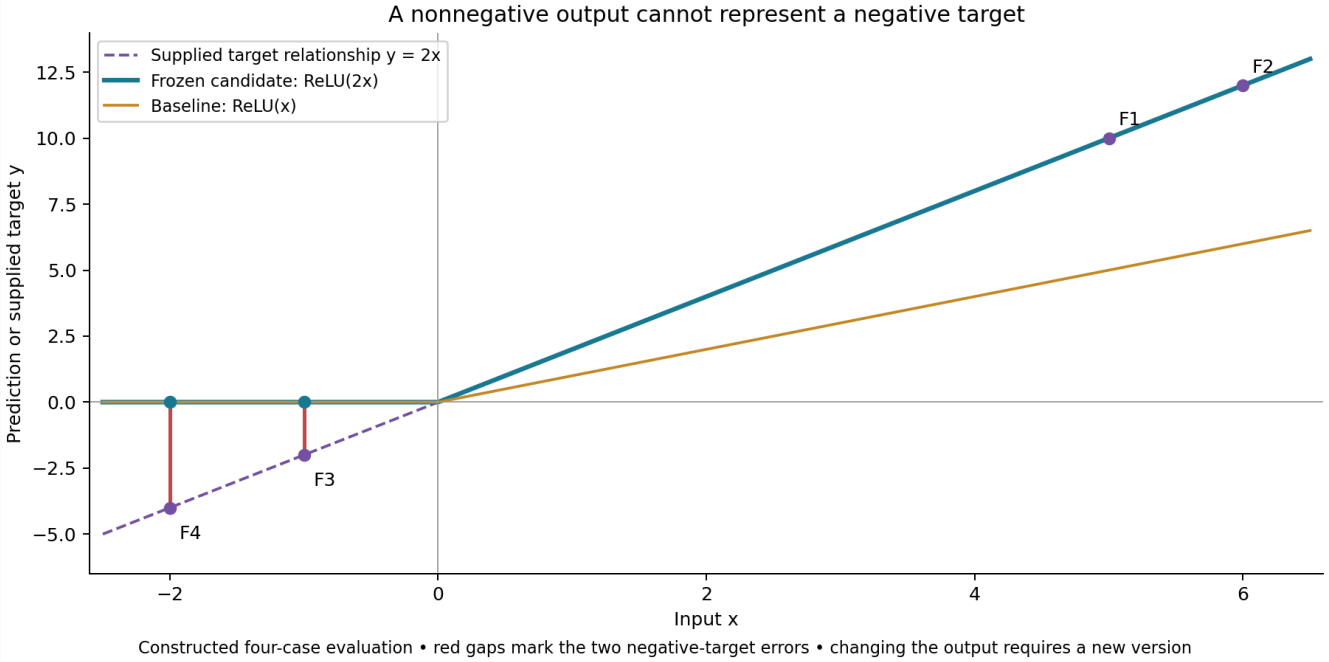
Baseline residual: Two cases

Baseline squared error: MSE 0

Candidate residual: Candidate negative subset

Candidate squared error: Two cases: MSE 10

Output-limit graphic and supported negative-case key



Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Frozen candidate a = 2	Final MSE = 5	Improves on baseline 20.25
Negative x cases	Predictions forced to zero	A structural limitation
Change output rule	Check revealed failures	New final evidence needed

A passed aggregate criterion and a known structural failure can both be true.

Guided: ReLU predicts 0, **residual** 6, squared error 36. Signed 2*x predicts -6, **residual** 0, squared error 0 on the revealed case. This supports representational feasibility, not independent generalisation.

Exit: original candidate MSE 5 versus baseline 20.25 on four constructed final rows; the changed architecture lacks new independent evaluation.

The ReLU **output** cannot be negative. When targets are negative, no choice of this scalar weight can produce the required negative **output**. More epochs do not remove that **output** constraint.

A proposed signed linear **output** a*x would represent negative targets, but it changes the architecture. It needs a new version, appropriate **fitting** and selection, and fresh reserved evaluation.

Running that proposal on F3 and F4 checks the revealed failures. It does not create new independent final **evidence**. Preserve the original test result and label later scores as **revision** or regression checks.

Do not replace the earlier failure report with a score from a new model on already inspected rows.

Classroom exposure and assessment record

Published final rows demonstrate evaluation arithmetic. Before an actual independent assessment, confirm that the learner has not inspected those inputs or targets during model choices; otherwise use fresh reserved cases. Assess correct evidence handling and explanation even when a model fails. Do not mark a model's success as proof of the learner's independent understanding.

Separate pictured slice audit and scope

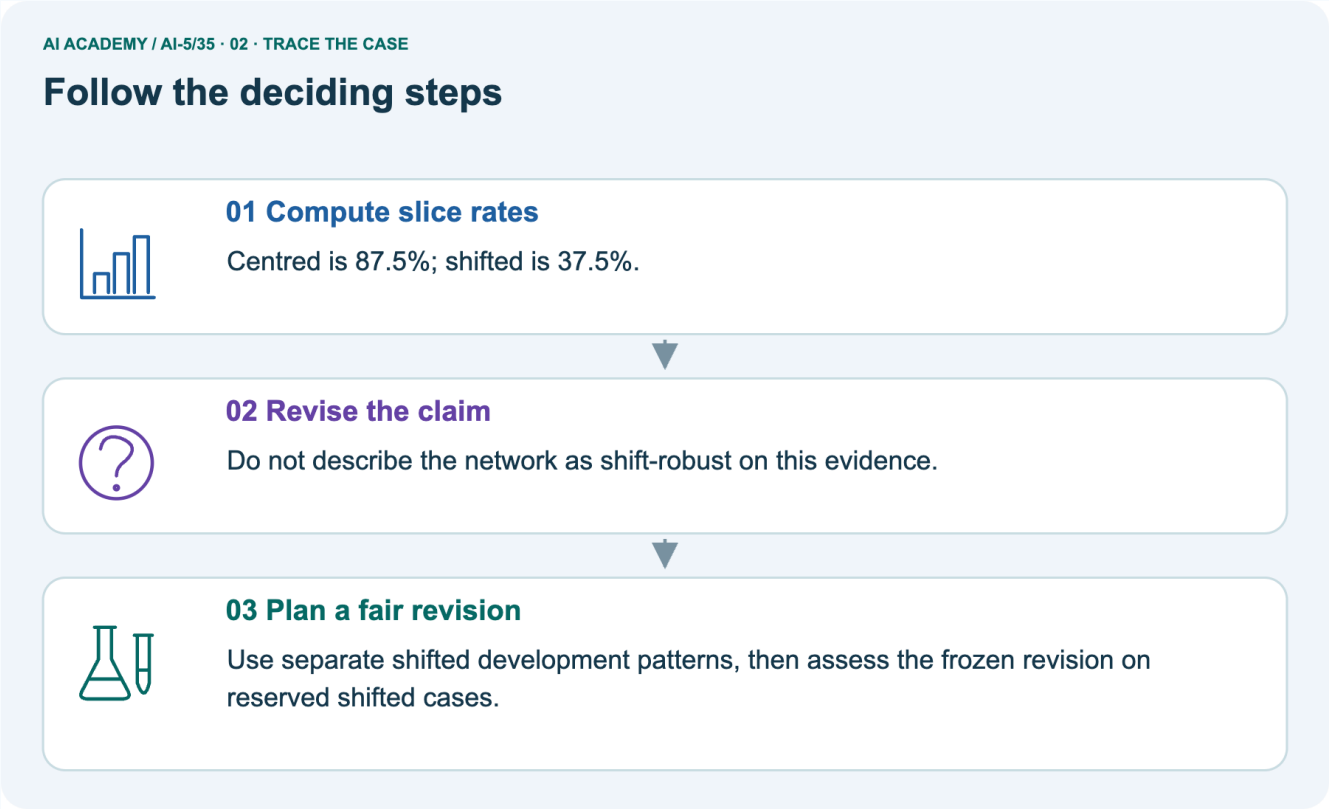
Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

Centred slice: 7/8 correct. Shifted slice: 3/8 correct. The network and label rubric are frozen.

The main experiment uses regression MSE; the picture is a separate supplied classification audit. Centred patterns score 7/8 = 87.5%, shifted patterns 3/8 = 37.5%. These observed counts do not prove the cause or estimate a stable future rate. Exact images and model parameters are not supplied. If all eight shifted evaluation patterns are used for a revision, rescoring them is exposed-case development evidence.

New reserved source groups are needed for an independent revised-model evaluation.

Picture trace and exposed-case revision key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute slice rates	Centred is 87.5%; shifted is 37.5%.
2. Revise the claim	Do not describe the network as shift-robust on this evidence.
3. Plan a fair revision	Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

Small slices may vary; this example diagnoses an observed weakness without inventing a population rate or cause.

Change: Train the **revision** on all eight failed-slice evaluation patterns and rescore them.

Prediction: Treat the score as exposed-case development, not independent **evidence** of shift robustness.

Reason: The evaluation records now influenced learning and model selection.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Frozen procedure	A recorded procedure fixed before final evaluation
Residual	Prediction minus observed target
Revision	A documented change informed by earlier evidence
Coverage limitation	A condition or group missing or poorly represented in the evidence

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Capstone limitations and revision

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/35

Capstone limitations and revision

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice rates

Centred is 87.5%; shifted is 37.5%.





2. Revise the claim

Do not describe the network as shift-robust on this evidence.





3. Plan a fair revision

Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases.

CHANGE ONE THING

Train the revision on all eight failed-slice evaluation patterns and rescore them.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Compute both overall MSE values and the candidate negative-**input** MSE. Apply the strict final criterion.

Teacher check: Teacher explanation: Baseline MSE 20.25; candidate MSE 5. Candidate negative-**input** MSE 10 over two rows. Candidate passes the lower-overall-MSE criterion but has a structural negative-**output** limitation.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Capstone limitations and **revision**. Proposed explanation: “Passing a criterion means no failures remain”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: An aggregate improvement can coexist with systematic subgroup errors.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate the slice rates and explain the claim **revision**. If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

The rates are 87.5% and 37.5%; the shift-robust claim is unsupported here. **Training** on those evaluation patterns makes a rerun development **evidence** on exposed cases, requiring fresh reserved cases for an independent check.

Changed-case reasoning: Treat the score as exposed-case development, not independent **evidence** of shift robustness. The evaluation records now influenced learning and model selection.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a frame with question, fixed settings, changed setting, data roles, raw **prediction**s, metric, conclusion and limitation. Allow a calculator and explicit squared-error columns.

Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and ask students to explain the data roles.
7 to 18 minutes	Read section 1 and complete W1 with a recorded reason.
18 to 32 minutes	Trace section 2 and complete W2; compare raw values before conclusions.
32 to 45 minutes	Explain the diagram and complete W3 in pairs; record any support.
Session 2 start to 8 minutes	Complete the guided response and discuss why the evidence has limits.
8 to 20 minutes	Read the remaining explanation and prepare a concise experiment record.
20 to 35 minutes	Collect W4 independently. Require calculations and a bounded claim.
35 to 45 minutes	Use W5 and the exit response for feedback. Administer the fresh retry after feedback without the earlier solution.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main baseline MSE 20.25, candidate 5, negative subset 10; aggregate criterion passes with a structural limitation. Independent frozen a3: baseline squares 16, 64, 9, 36→31.25; candidate 0, 0, 9, 36→11.25; negative MSE 22.5. Both improve only on the stated four-case comparison. No claimed new revision performance, causal image explanation or universal robustness.

Diagnose the current misconceptions

Passing a criterion means no failures remain: An aggregate improvement can coexist with systematic subgroup errors.

Training longer fixes any **output** constraint: A nonnegative **output** cannot express a negative target.

A repaired test score is a new final test: The revealed cases informed the repair and now serve as development or regression checks.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12),(-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion.

Teacher explanation: Baseline squared errors 16,64,9,36 total 125, MSE 31.25. Candidate squared errors 0,0,9,36 total 45, MSE 11.25. Negative-**input** MSE 22.5 over two rows.

Candidate improves overall on these four constructed cases but still cannot represent negative **output**s.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace final predictions

For F1–F4 calculate candidate **residual**s and squared errors.

Teacher explanation: **Residual**s 0,0,2,4; squared errors 0,0,4,16. The ReLU **prediction**s for both negative **input**s are zero.

W2 Compare with the baseline

Compute both overall MSE values and the candidate negative-**input** MSE. Apply the strict final criterion.

Teacher explanation: Baseline MSE 20.25; candidate MSE 5. Candidate negative-**input** MSE 10 over two rows. Candidate passes the lower-overall-MSE criterion but has a structural negative-**output** limitation.

W3 Explain the architecture limit

Why cannot extra **training** alone make this ReLU **output** negative? Name a proposed change and the **evidence** it would require.

Teacher explanation: max(0,a*x) is always nonnegative, regardless of a.

A signed linear **output** can represent negative values; treat it as a changed architecture and run a recorded **fitting**/selection procedure with fresh reserved final **evidence**.

W5 Label a revision honestly

After inspecting F1–F4, a student changes to **prediction** 2*x and reports “new final MSE 0”. Explain the accurate label and the next evaluation step.

Teacher explanation: The score is a **revision** check on already inspected final rows. Preserve the original final MSE 5 and the architectural change.

Freeze the revised **training** and selection procedure and evaluate on fresh reserved cases; also retain regression checks.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: ReLU predicts 0, **residual** 6, squared error 36. Signed 2*x predicts -6, **residual** 0, squared error 0 on the revealed case. This supports representational feasibility, not independent generalisation.

Exit: original candidate MSE 5 versus baseline 20.25 on four constructed final rows; the changed architecture lacks new independent evaluation.

W1 Trace final predictions

W2 Compare with the baseline

W3 Explain the architecture limit

W4 Evaluate new evidence independently

A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12),(-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion.
</details>

Teacher explanation: Baseline squared errors 16,64,9,36 total 125, MSE 31.25. Candidate squared errors 0,0,9,36 total 45, MSE 11.25. Negative-**input** MSE 22.5 over two rows.

Candidate improves overall on these four constructed cases but still cannot represent negative **output**s.

W5 Label a revision honestly

Independent assessment and fresh retry

W4 requires overall MSE 31.25 versus 11.25 and negative-**input** MSE 22.5, with a four-case limit. Fresh retry: a = 4, baseline 1, final rows (2,8),(4,16),(-1,-4),(-2,-8). Baseline MSE 65; candidate MSE 20; negative-**input** MSE 40.

Require an aggregate improvement claim alongside the structural failure.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12),(-1,-3),(-2,-6).

Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Provide a frame with question, fixed settings, changed setting, data roles, raw **prediction**s, metric, conclusion and limitation. Allow a calculator and explicit squared-error columns.

Extension: implement the arithmetic locally, preserve the program version, and explain why a successful replay adds no independent test cases.

Week 36 defends the experiment using preserved plans, results, limitations and an honest account of **revision**.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Delayed return and support record

Suggested return points in this level: ai-5/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Evaluate a frozen candidate and baseline on final rows

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Compute both overall MSE values and the candidate negative-**input** MSE. Apply the strict final criterion.

Reasoned practice answer: Teacher explanation: Baseline MSE 20.25; candidate MSE 5. Candidate negative-**input** MSE 10 over two rows.

Candidate passes the lower-overall-MSE criterion but has a structural negative-**output** limitation.

Check every learner before discussion: Use the worked case for Capstone limitations and **revision**. Proposed explanation: “Passing a criterion means no failures remain”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: An aggregate improvement can coexist with systematic subgroup errors.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Train the **revision** on all eight failed-slice evaluation patterns and rescore them. Require a **prediction** and a reason before comparing.

Expected result: Treat the score as exposed-case development, not independent **evidence** of shift robustness. Reason: The evaluation records now influenced learning and model selection.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

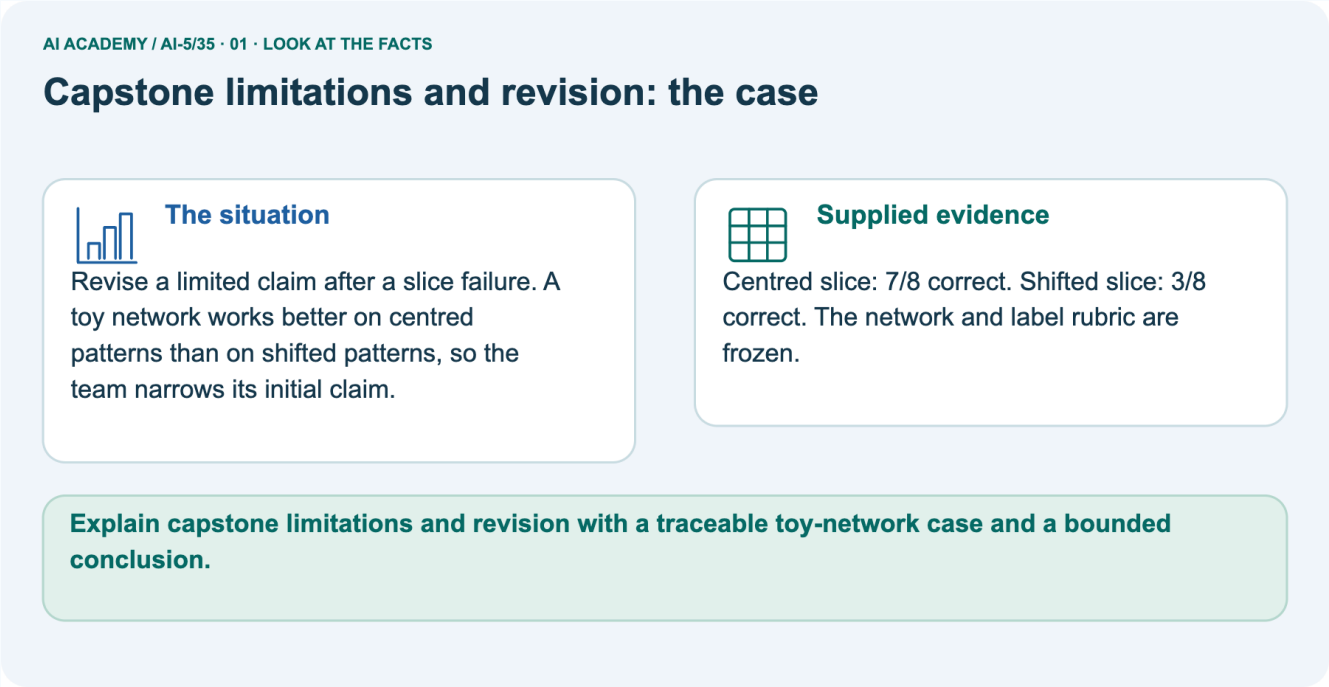
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Capstone limitations and revision: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

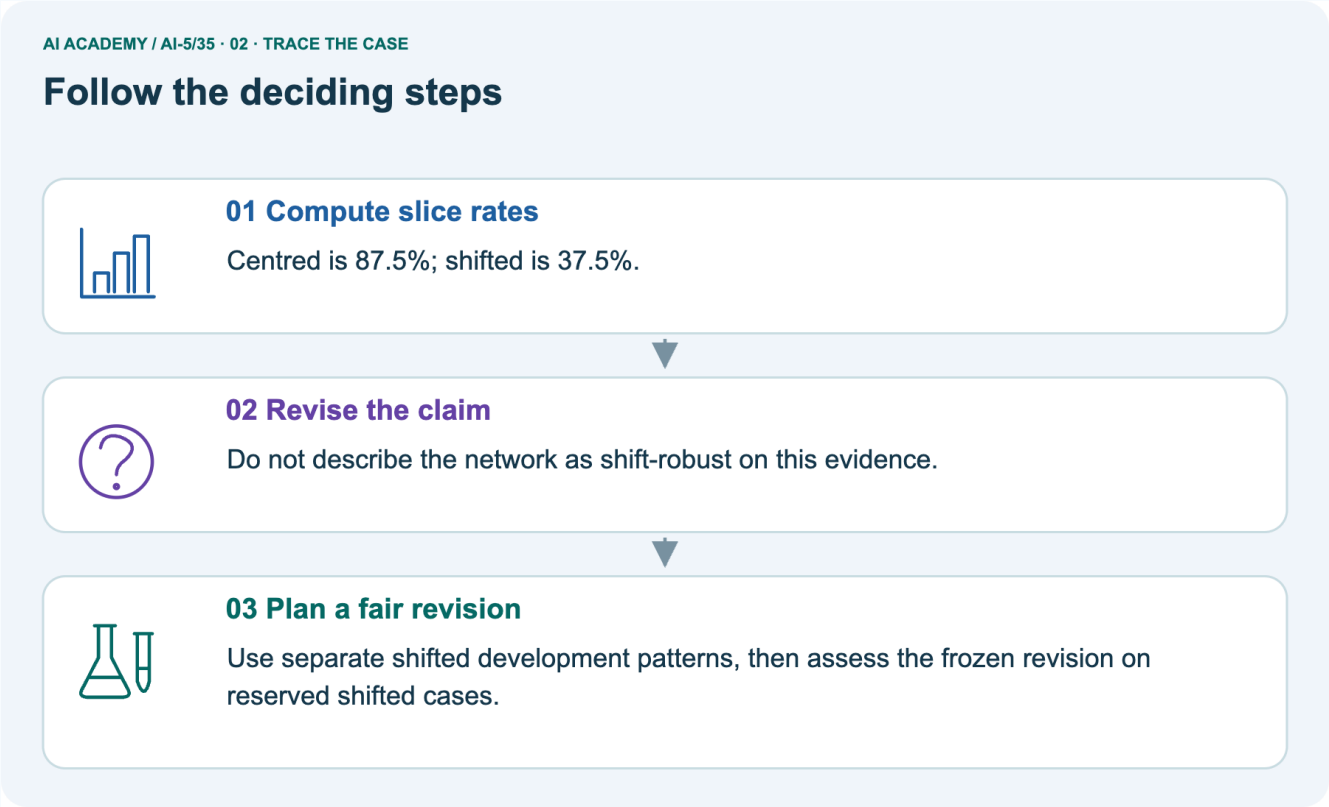
Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim. Supplied facts: Centred slice: 7/8 correct.

Shifted slice: 3/8 correct. The network and label rubric are frozen.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



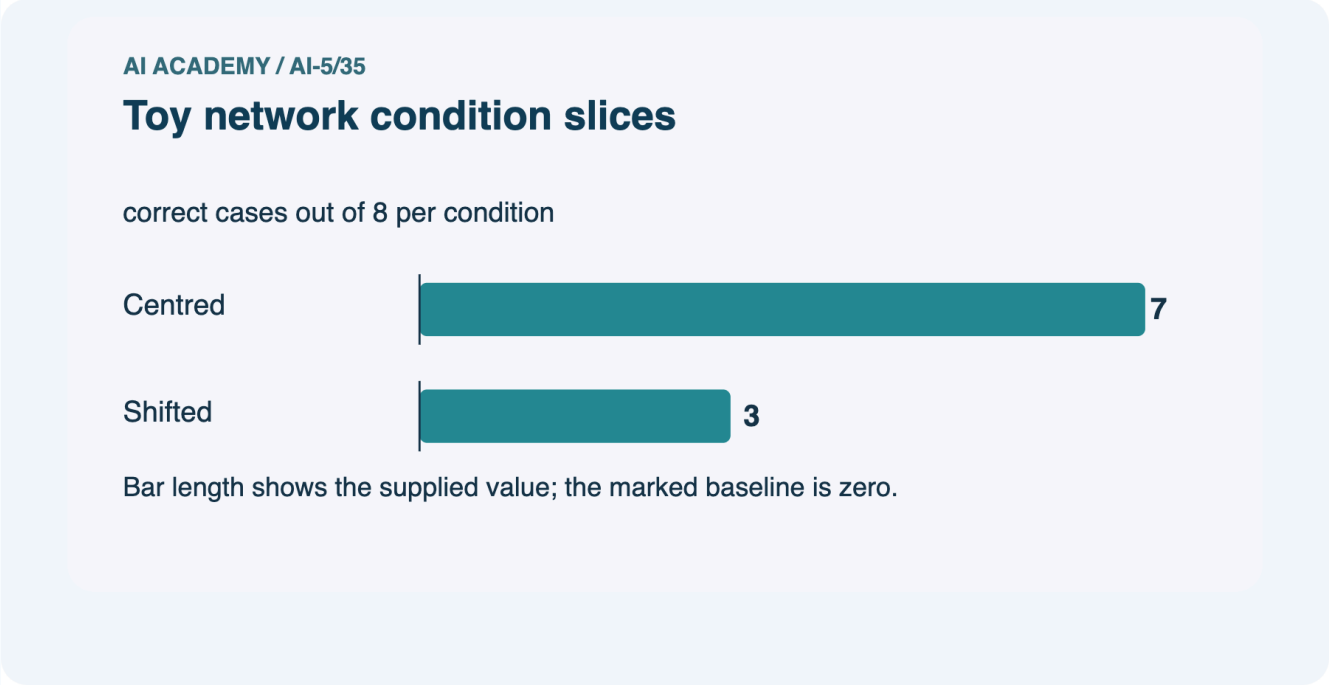
Follow the labelled picture with the matching explanation. Open the vector diagram

Compute slice rates: Centred is 87.5%; shifted is 37.5%. → Revise the claim: Do not describe the network as shift-robust on this **evidence**. → Plan a fair **revision**: Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Toy network condition slices



Follow the labelled picture with the matching explanation. Open the vector diagram

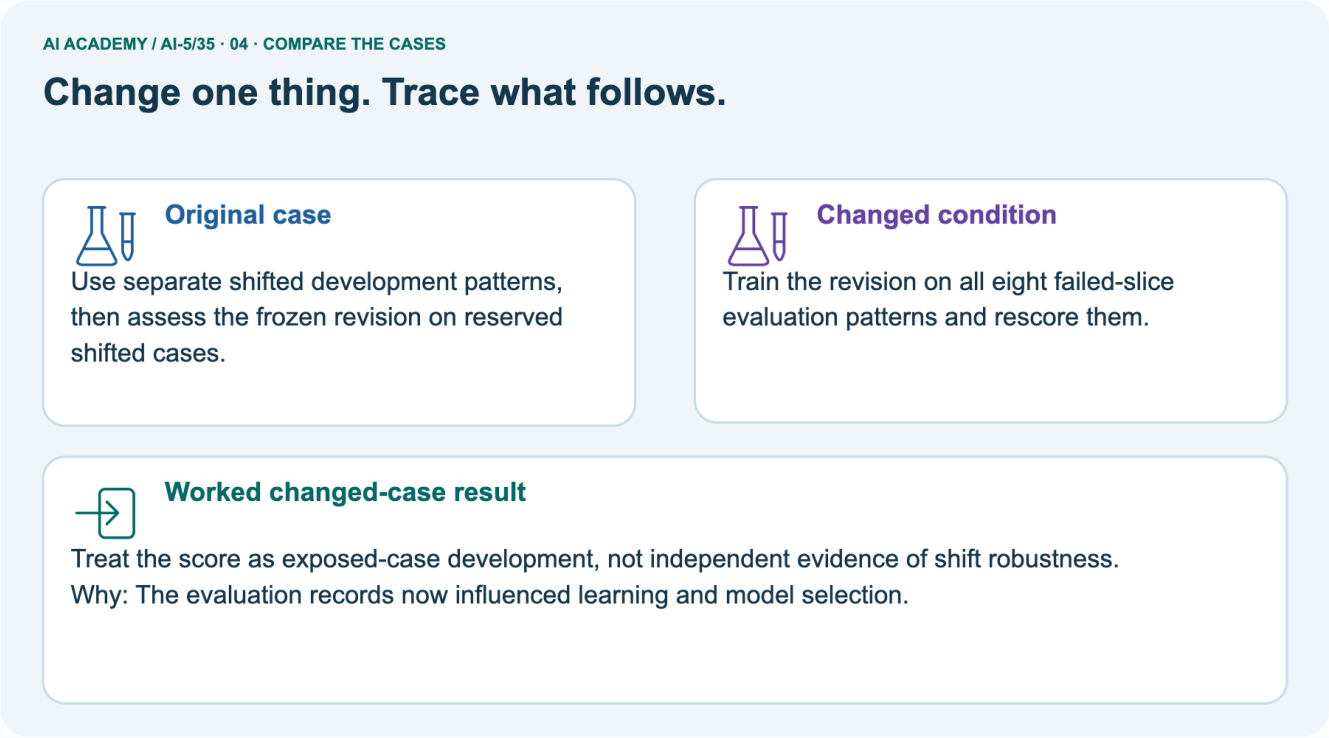
Toy network condition slices. Units: correct cases out of 8 per condition. Centred: 7. Shifted: 3.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases. Changed condition: Train the **revision** on all eight failed-slice evaluation patterns and rescore them.

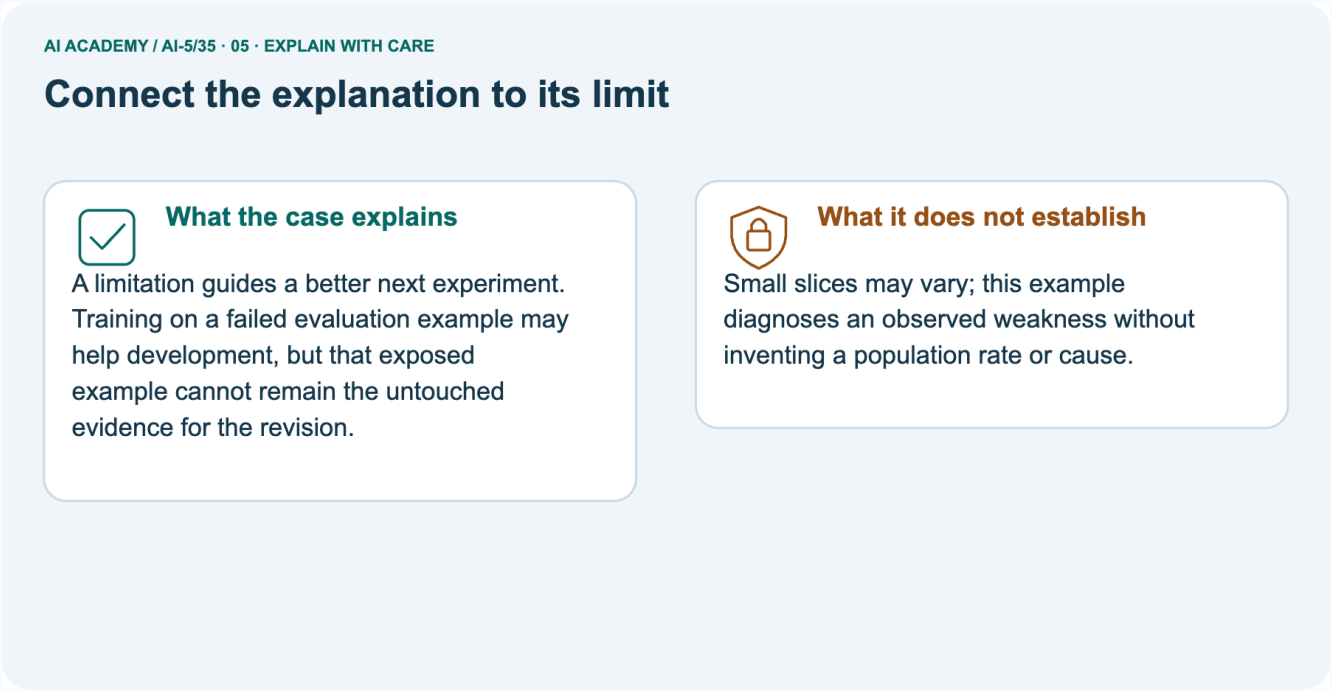
Expected result: Treat the score as exposed-case development, not independent **evidence** of shift robustness. Reason: The evaluation records now influenced learning and model selection.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The evaluation records now influenced learning and model selection.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

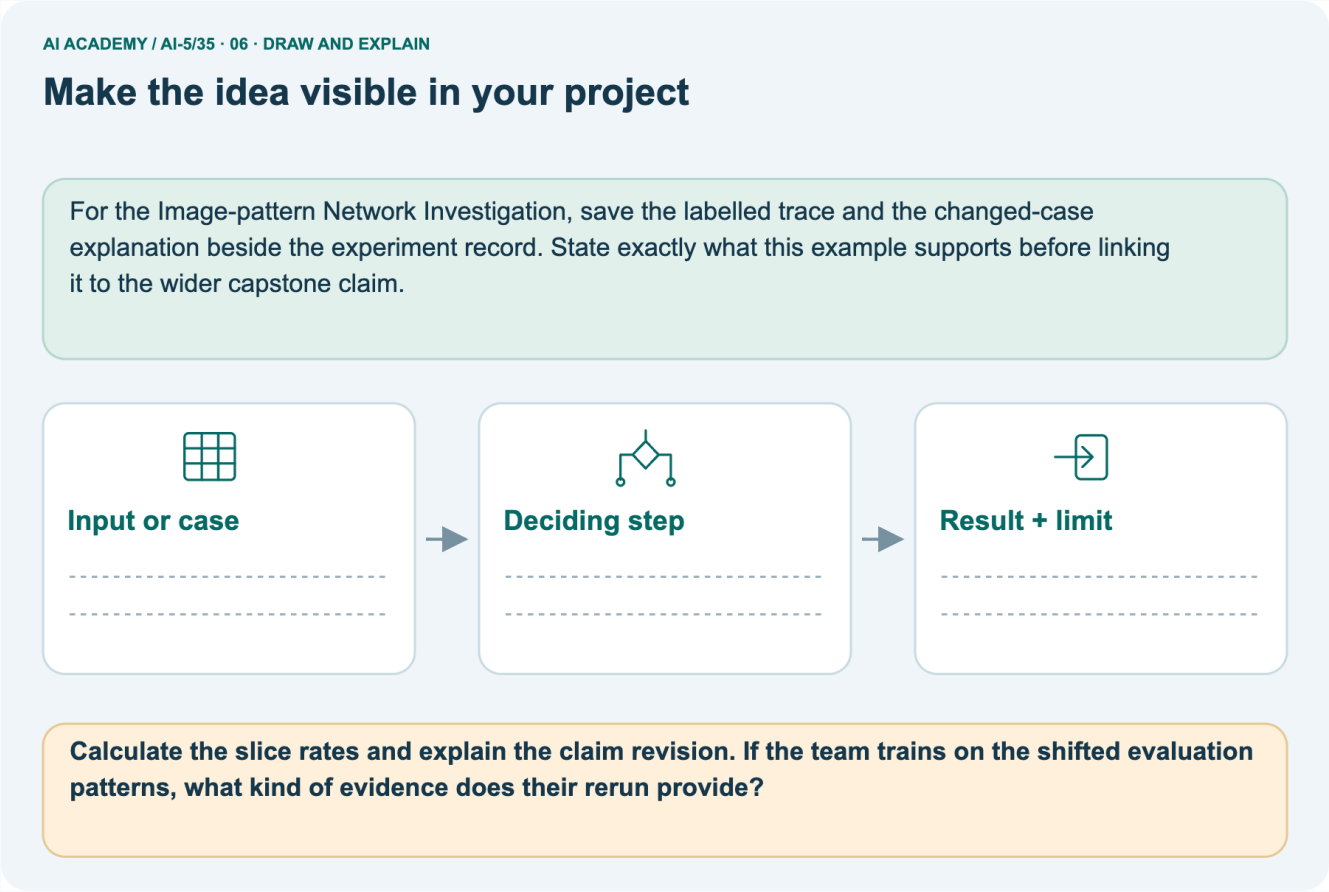
Limit: Small slices may vary; this example diagnoses an observed weakness without inventing a population rate or cause.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Small slices may vary; this example diagnoses an observed weakness without inventing a population rate or cause.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the slice rates and explain the claim **revision**.

If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The rates are 87.5% and 37.5%; the shift-robust claim is unsupported here.

Training on those evaluation patterns makes a rerun development **evidence** on exposed cases, requiring fresh reserved cases for an independent check.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Capstone limitations and revision: the case

02. Follow the deciding steps

03. Toy network condition slices

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.